

Introduction

How does the brain encode and stabilize a new memory or activity pattern in a noisy environment? Although current prevailing theories suggest that memories are encoded in patterns of synaptic weights within brain networks, the mechanisms by which these patterns are stabilized and reliably accessed are not well understood. Individual synapses are unreliable, and their weights and responses change over time.

Recent studies suggest that network activity patterns could be stabilized in rate based neural networks via complex plasticity mechanisms. However, it is still unclear whether the same mechanisms could produce stable memories in spiking neural networks and whether these rules are biologically plausible.

In our work, we are developing a biologically inspired spiking neural network that can learn and store multiple memories despite the presence of noise.

Methodology

Izhikevich Model of Spiking Neurons

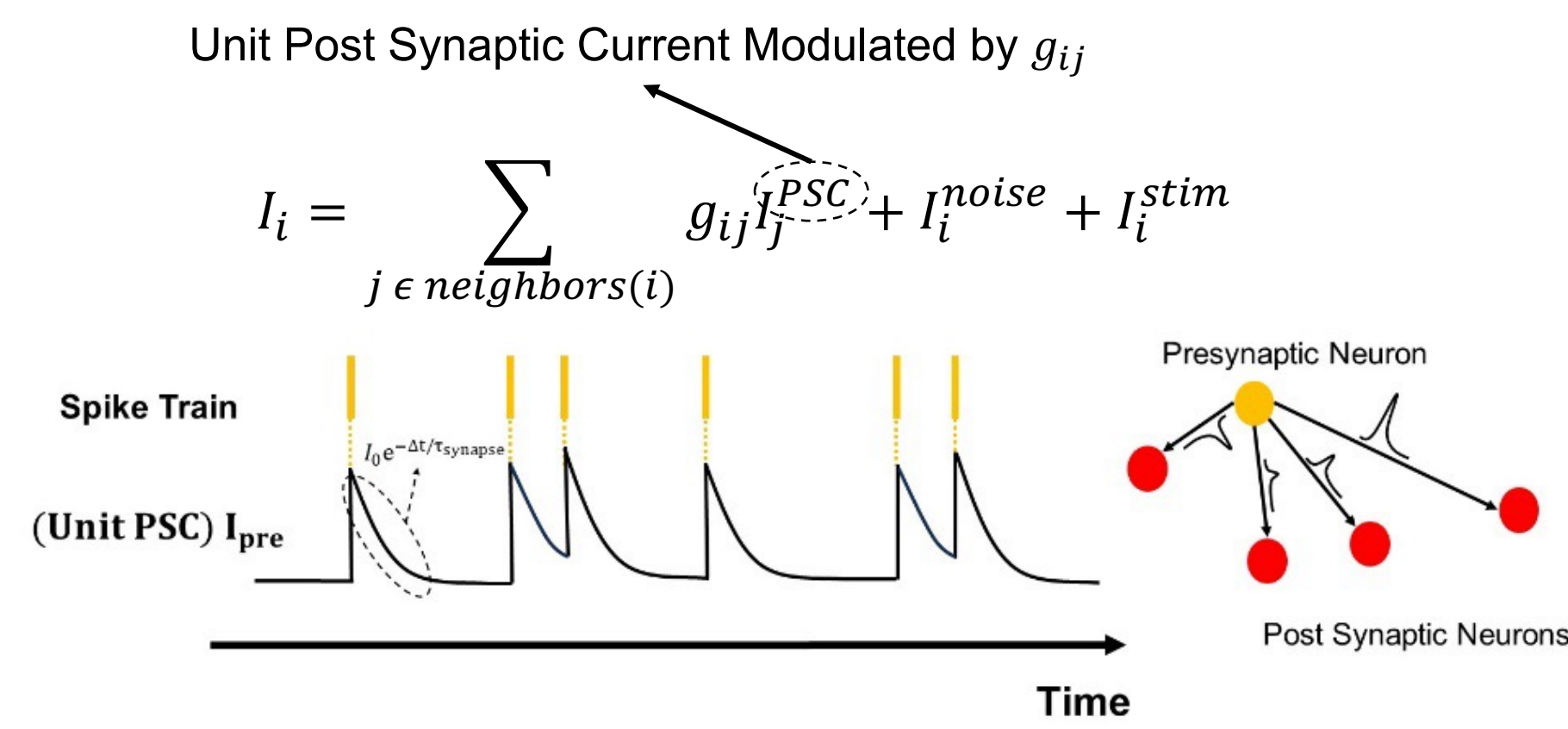
$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I$$

$$\frac{du}{dt} = a(bv - u)$$

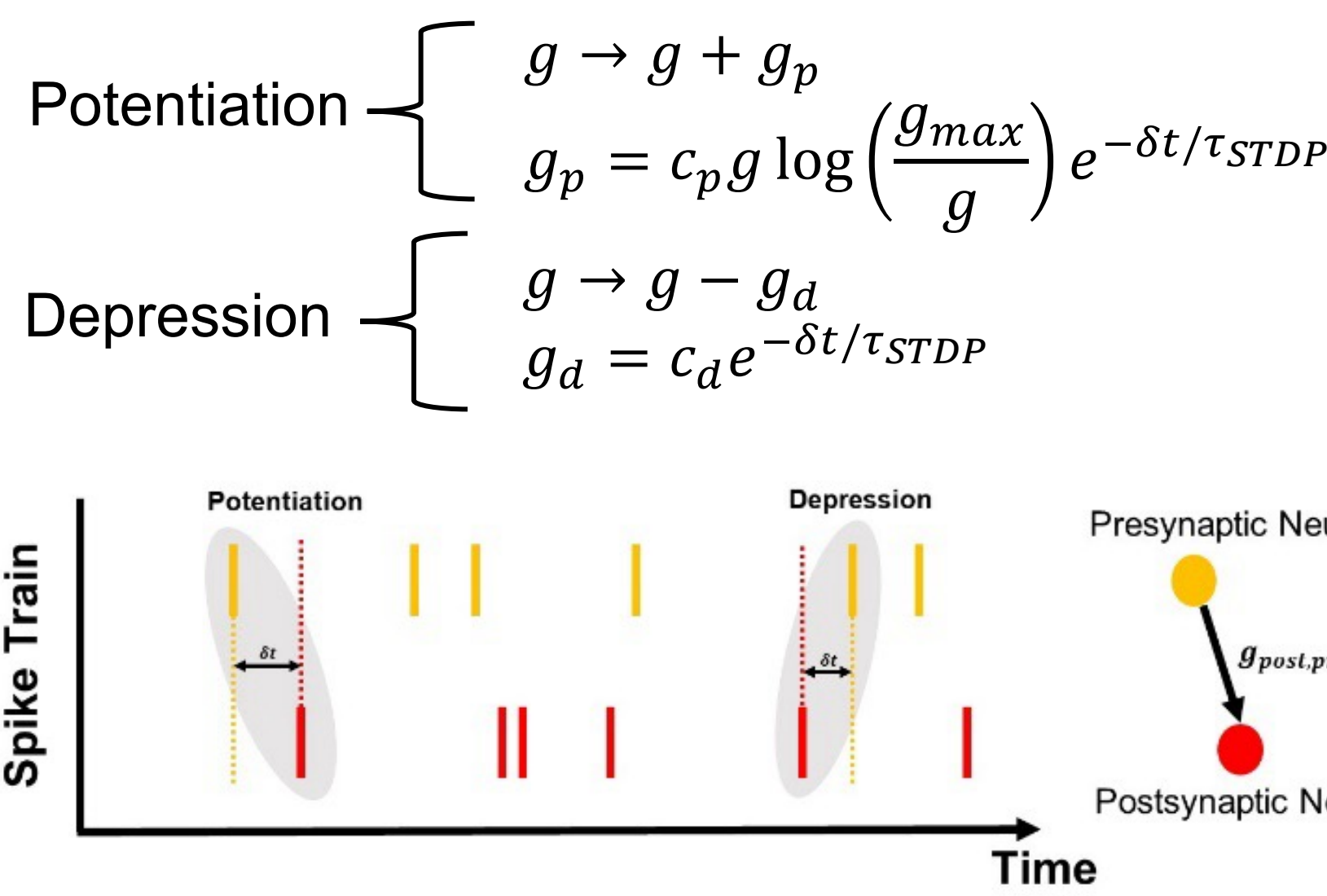
Neuron Input Current

if $v \geq 30 \text{ mV}$, then $\begin{cases} v \rightarrow c \\ u \rightarrow u + d \end{cases}$

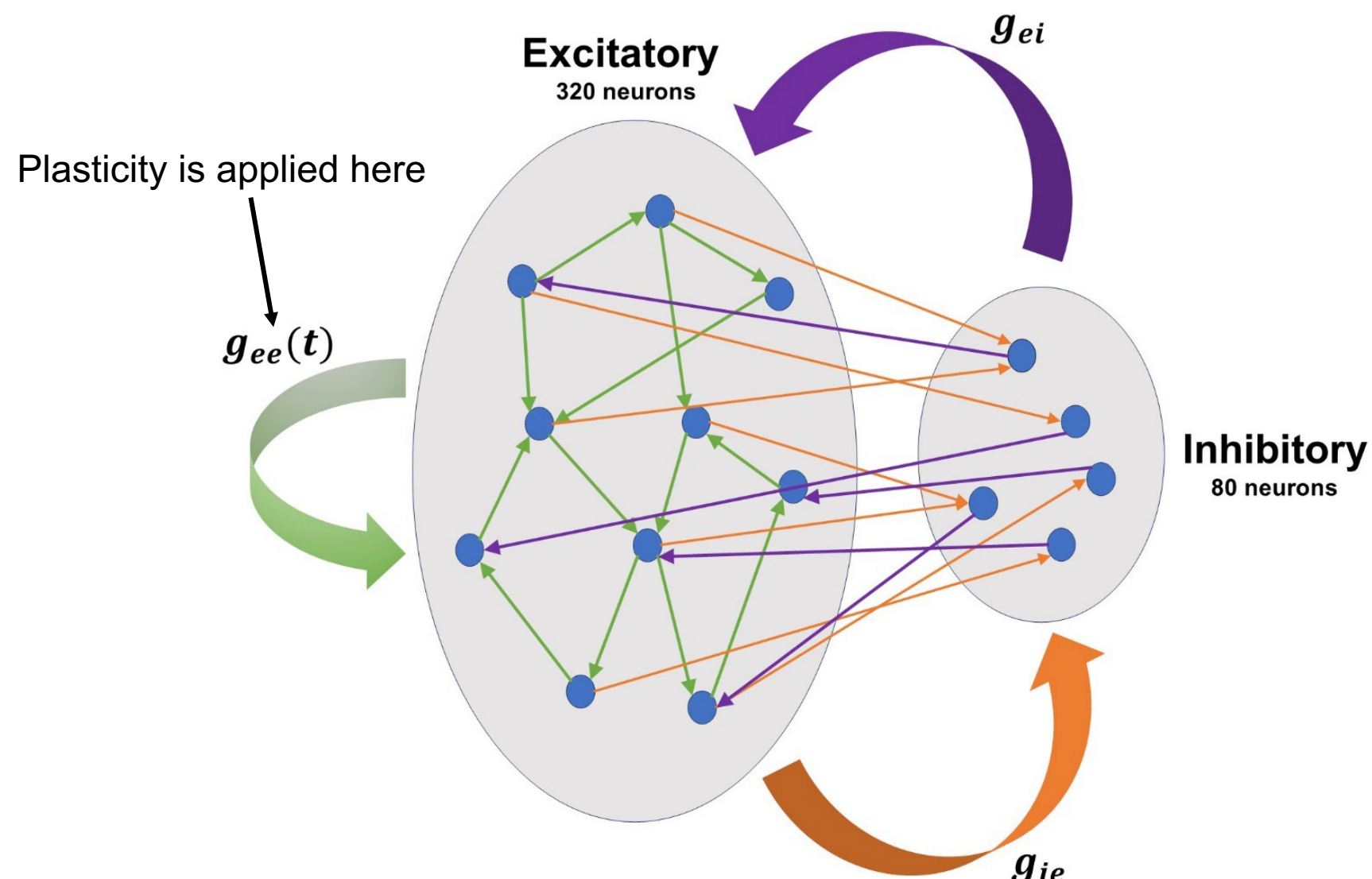
Input Current Model



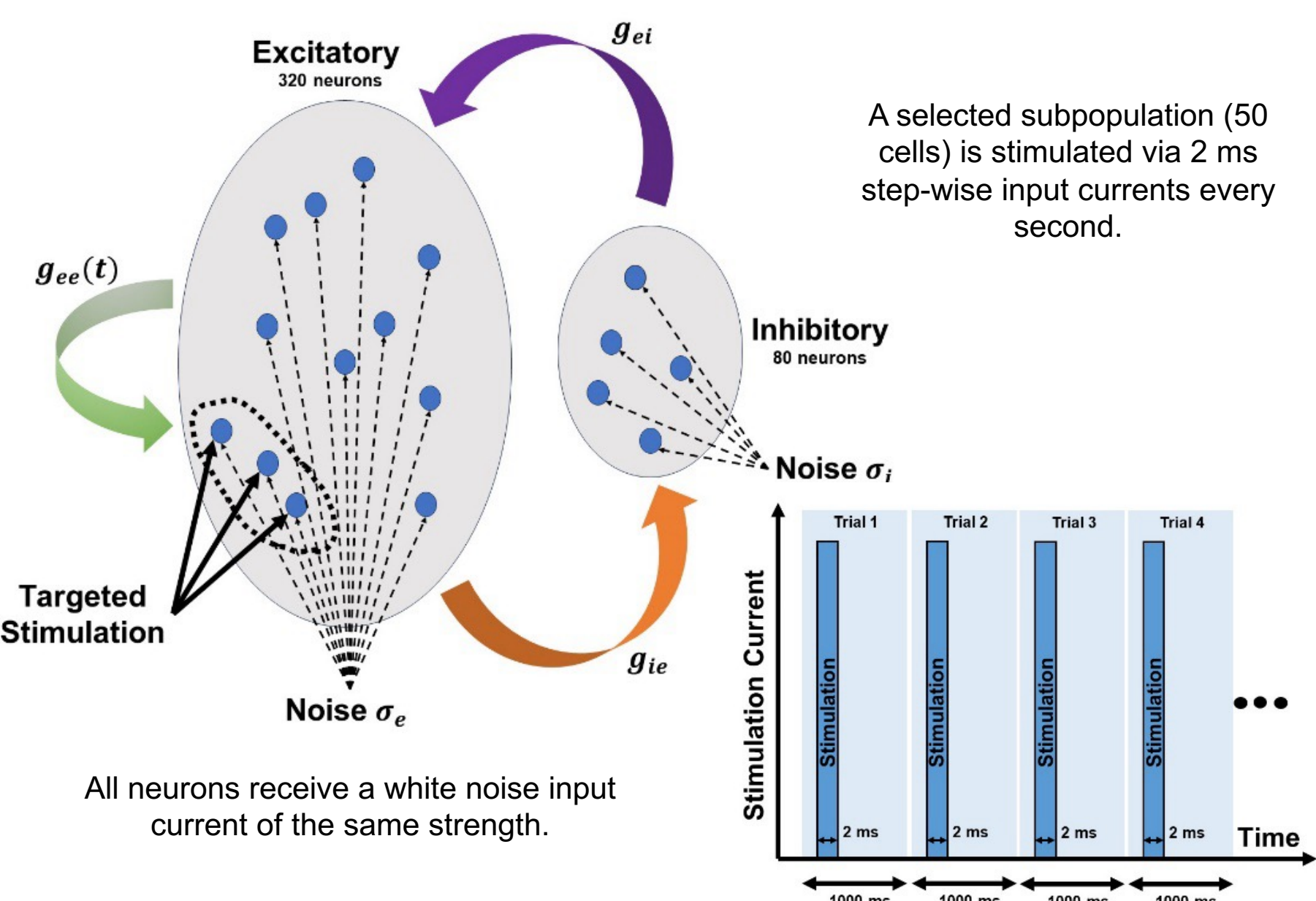
Hebbian Spike Time Dependent Plasticity Rule



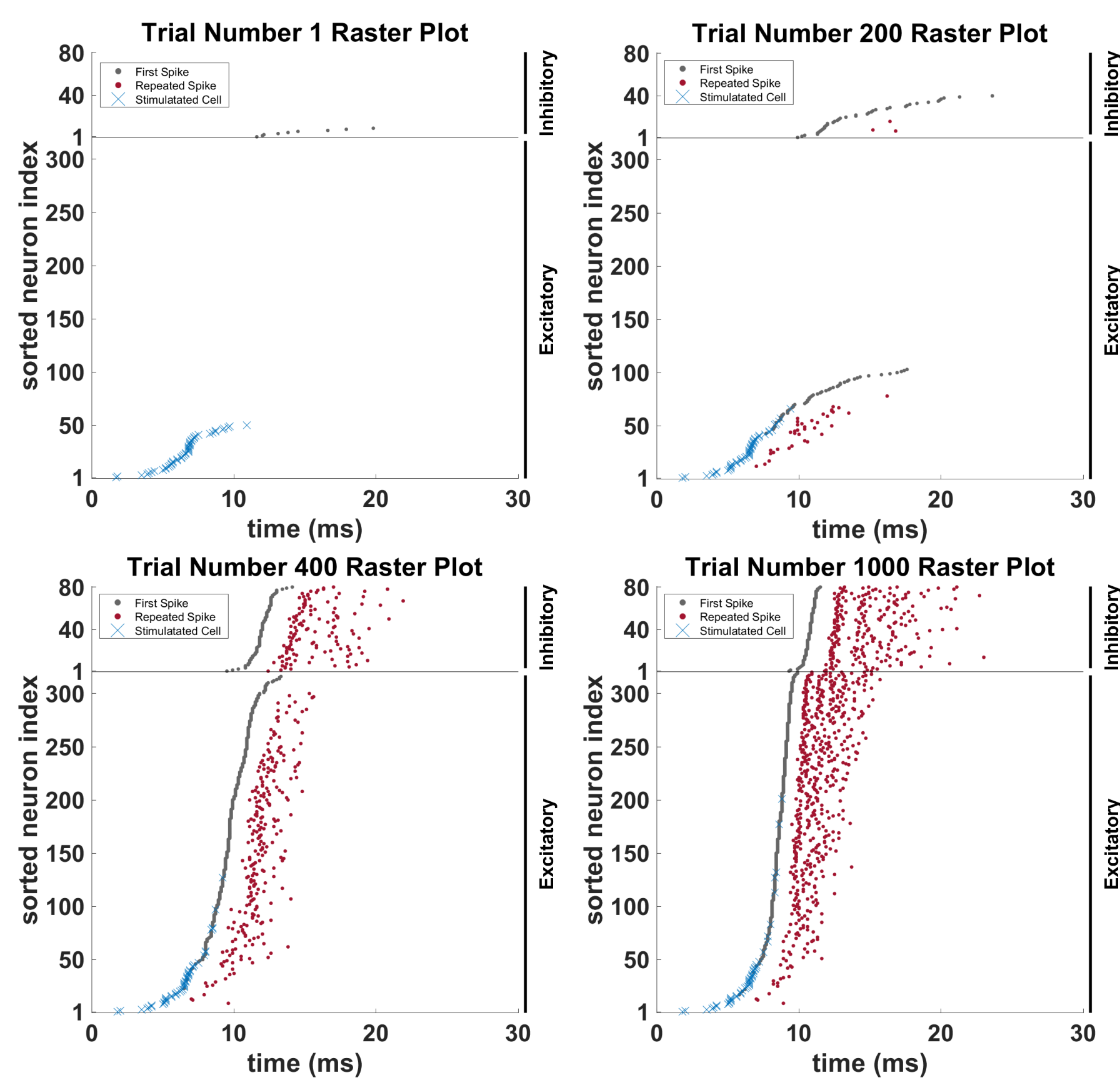
Network Architecture



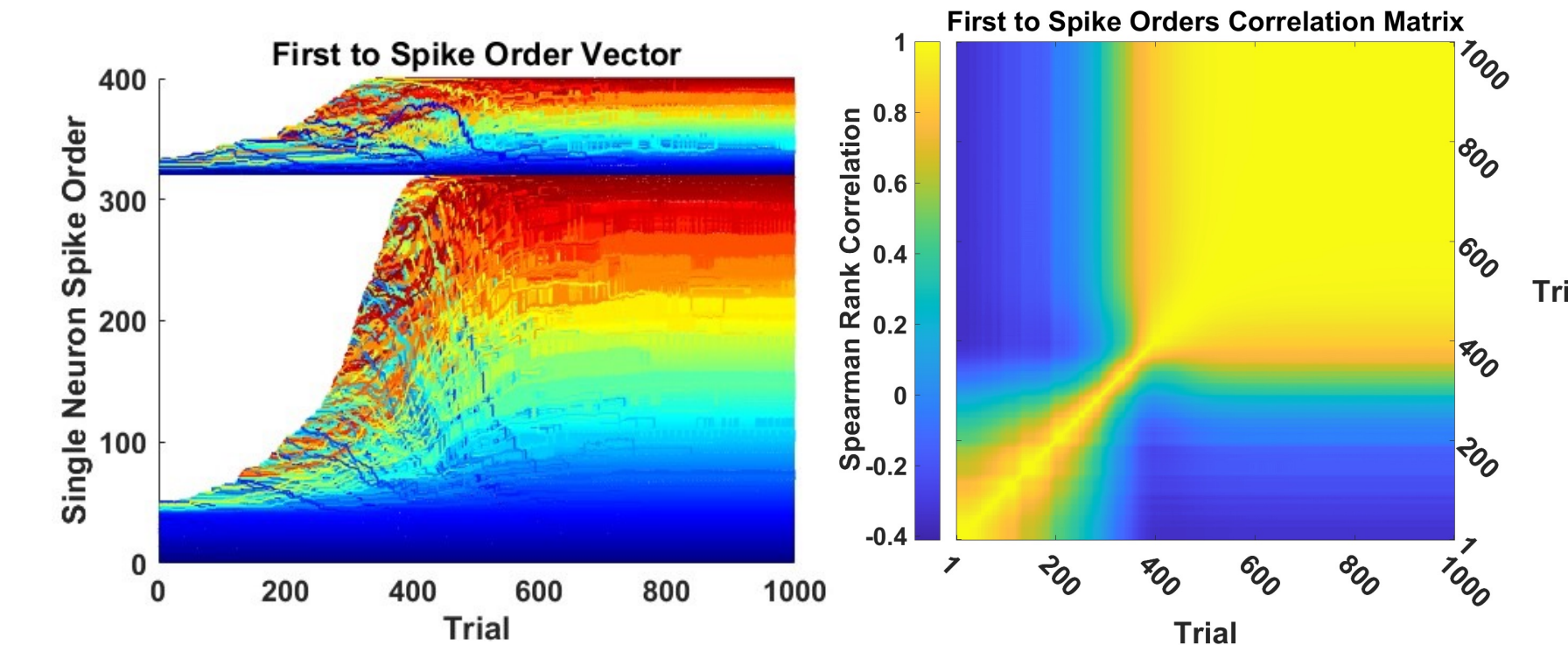
Stimulation and Noise



Imprinting a Single Memory

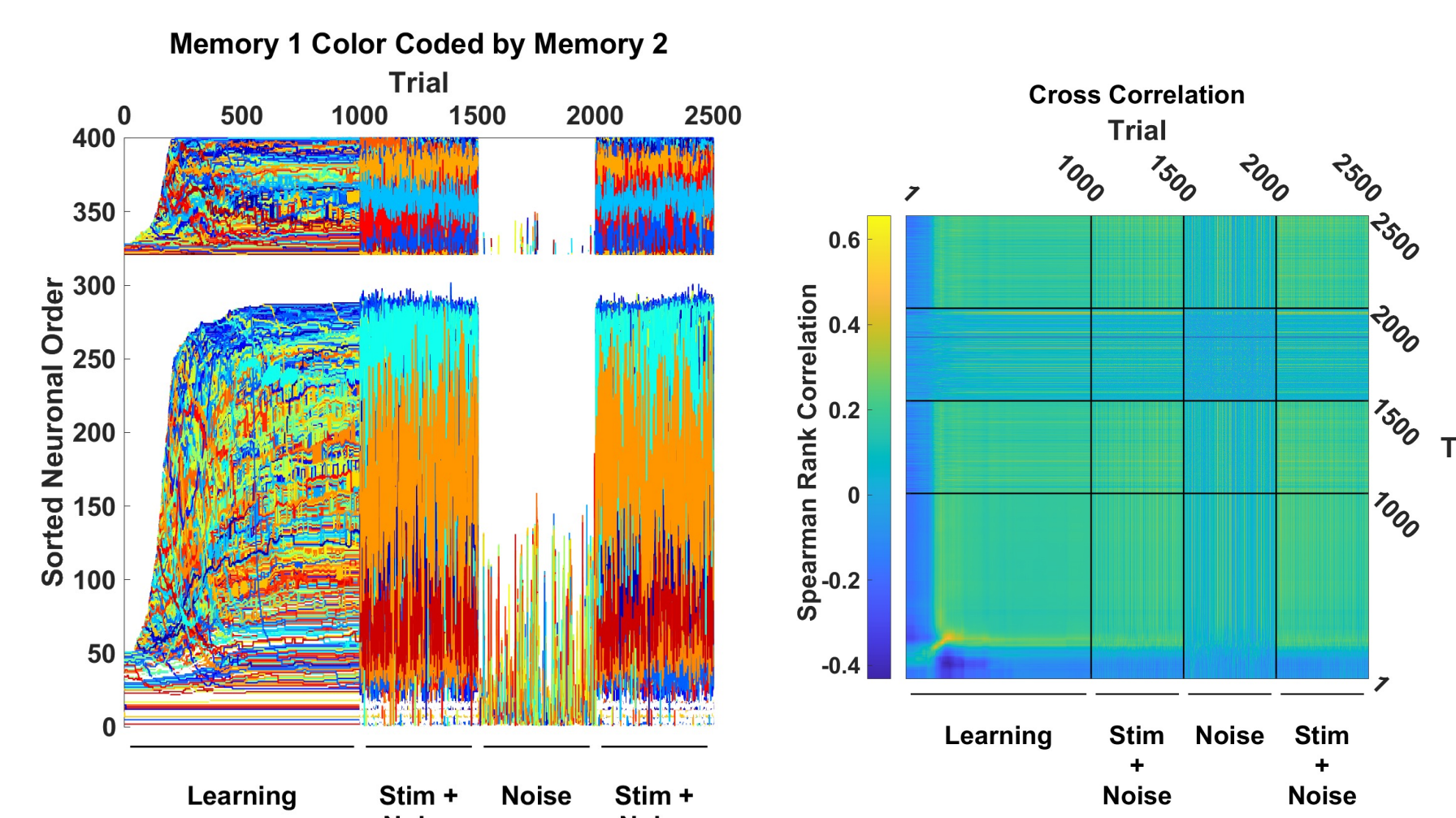
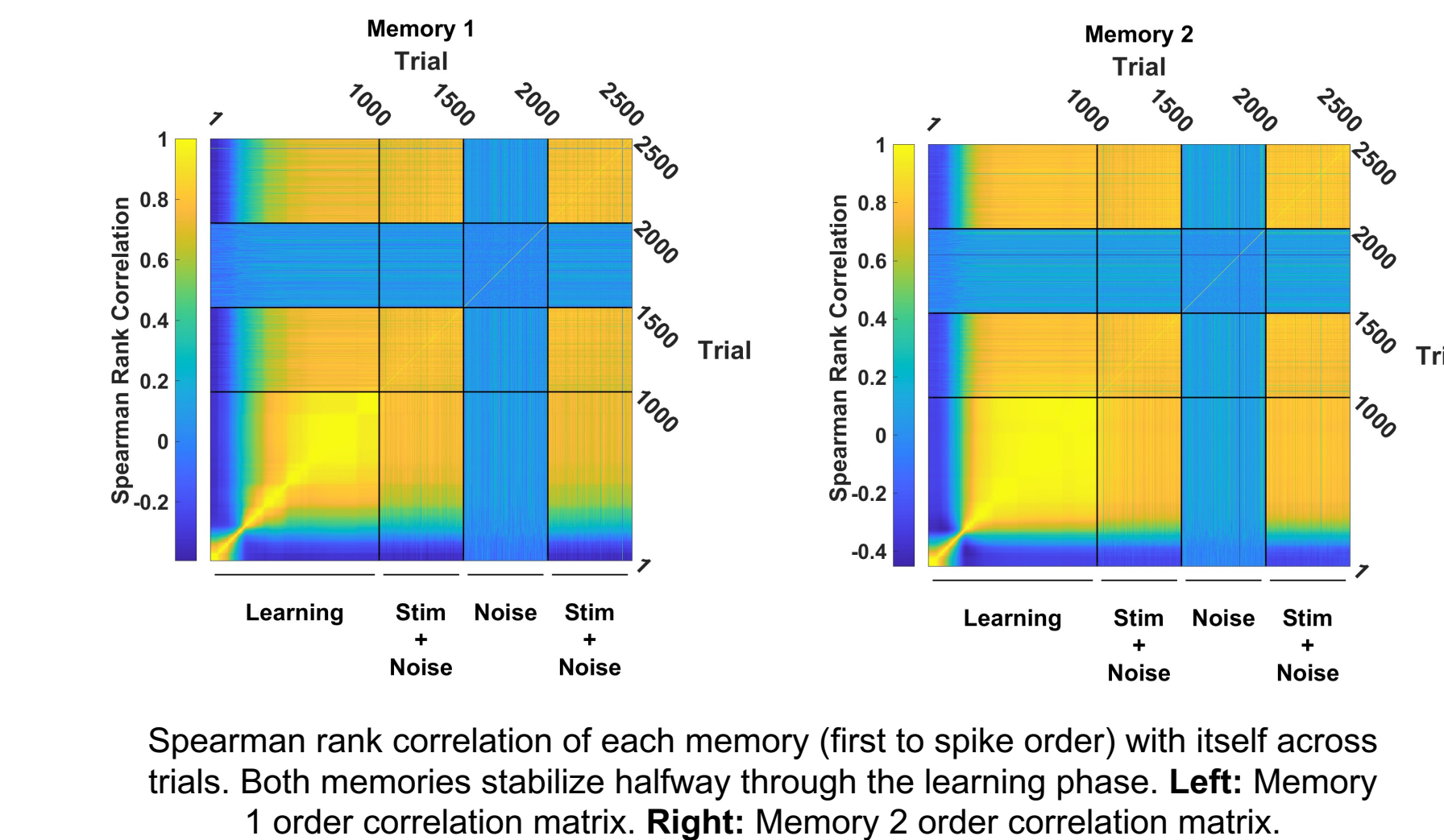
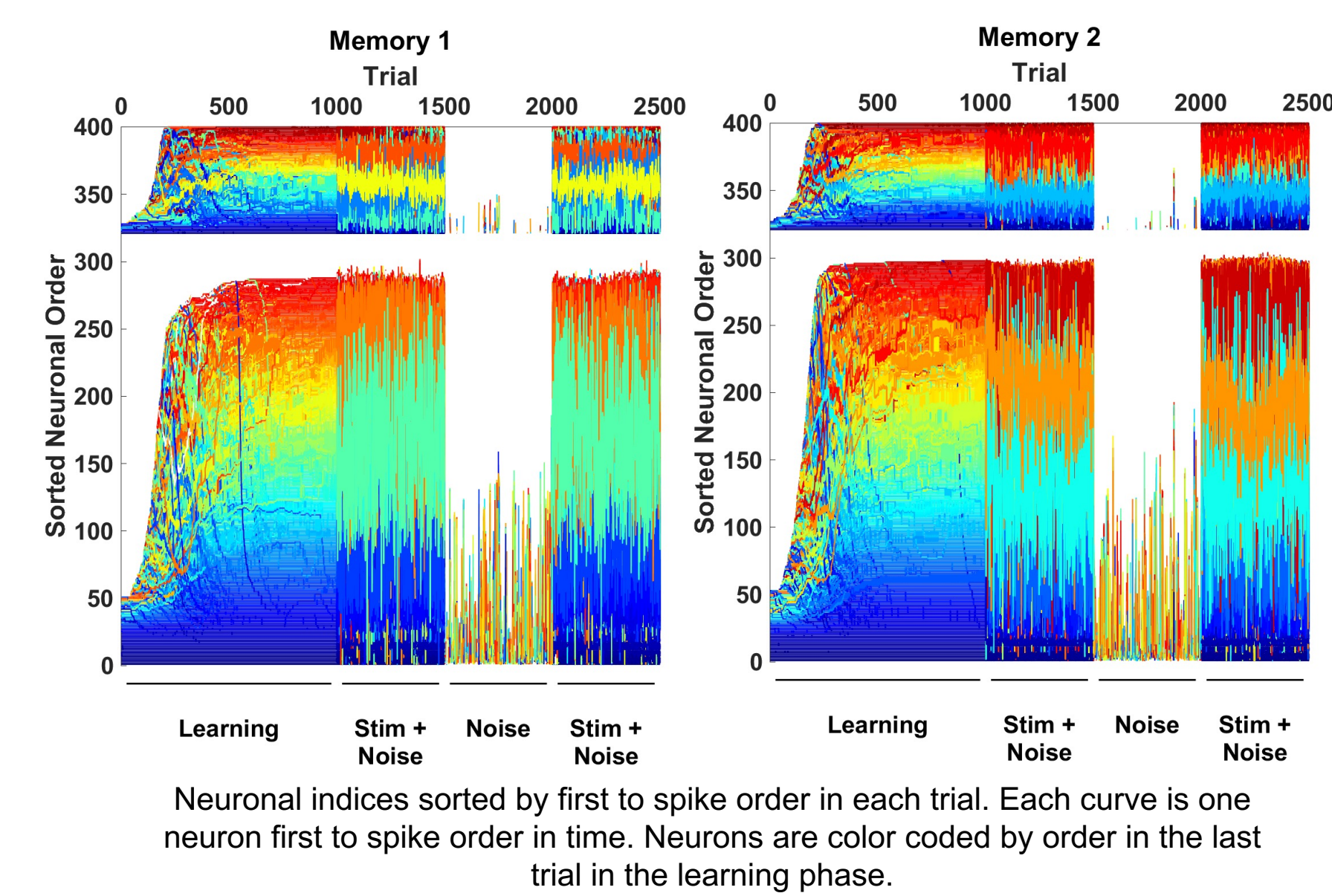
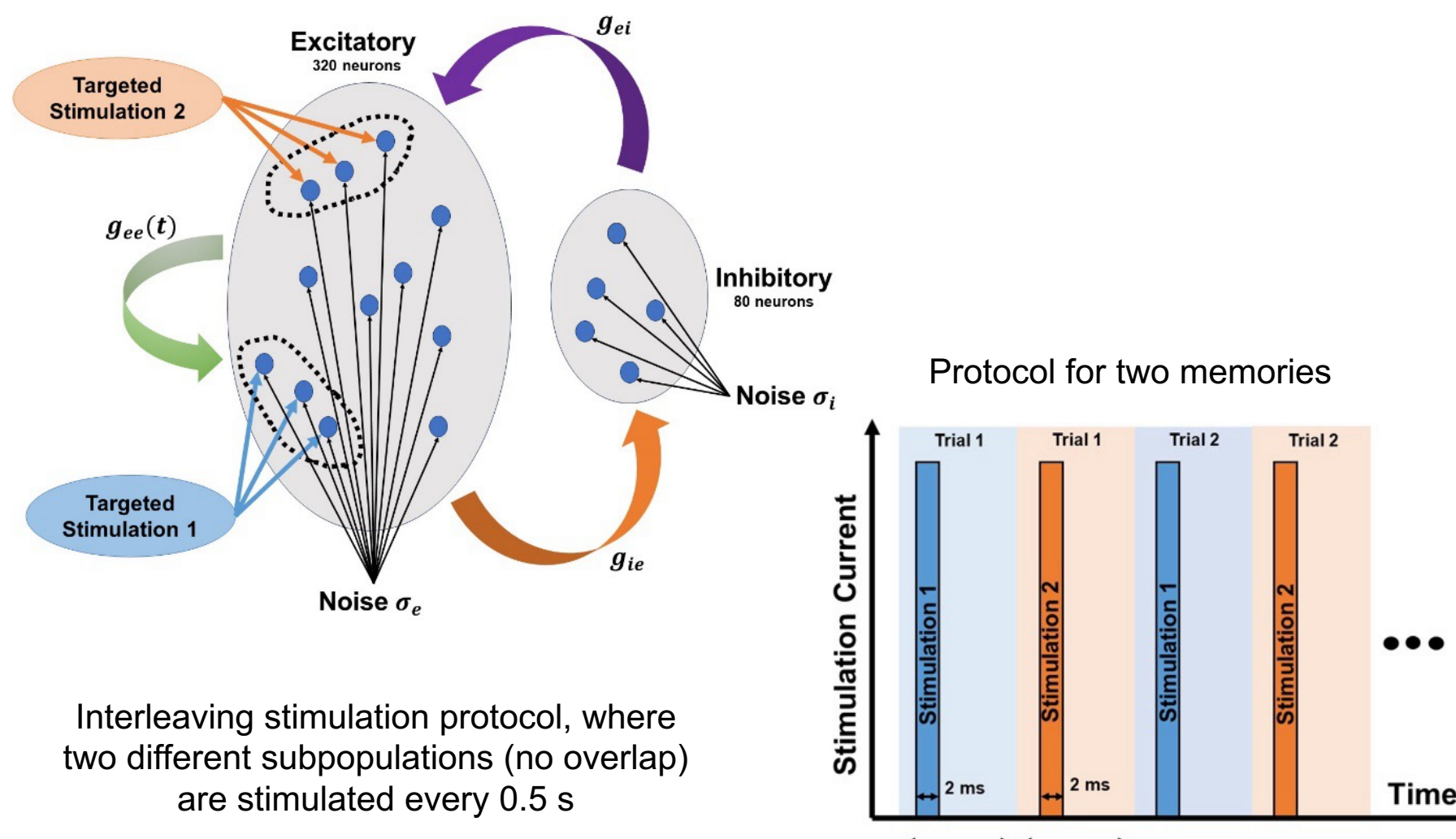


Raster plots of the network response to the stimulation along four different trials throughout the stimulation protocol. By trial 400 the sequence has stabilized, and its order barely changes until the end of the stimulation protocol (trial 1000)

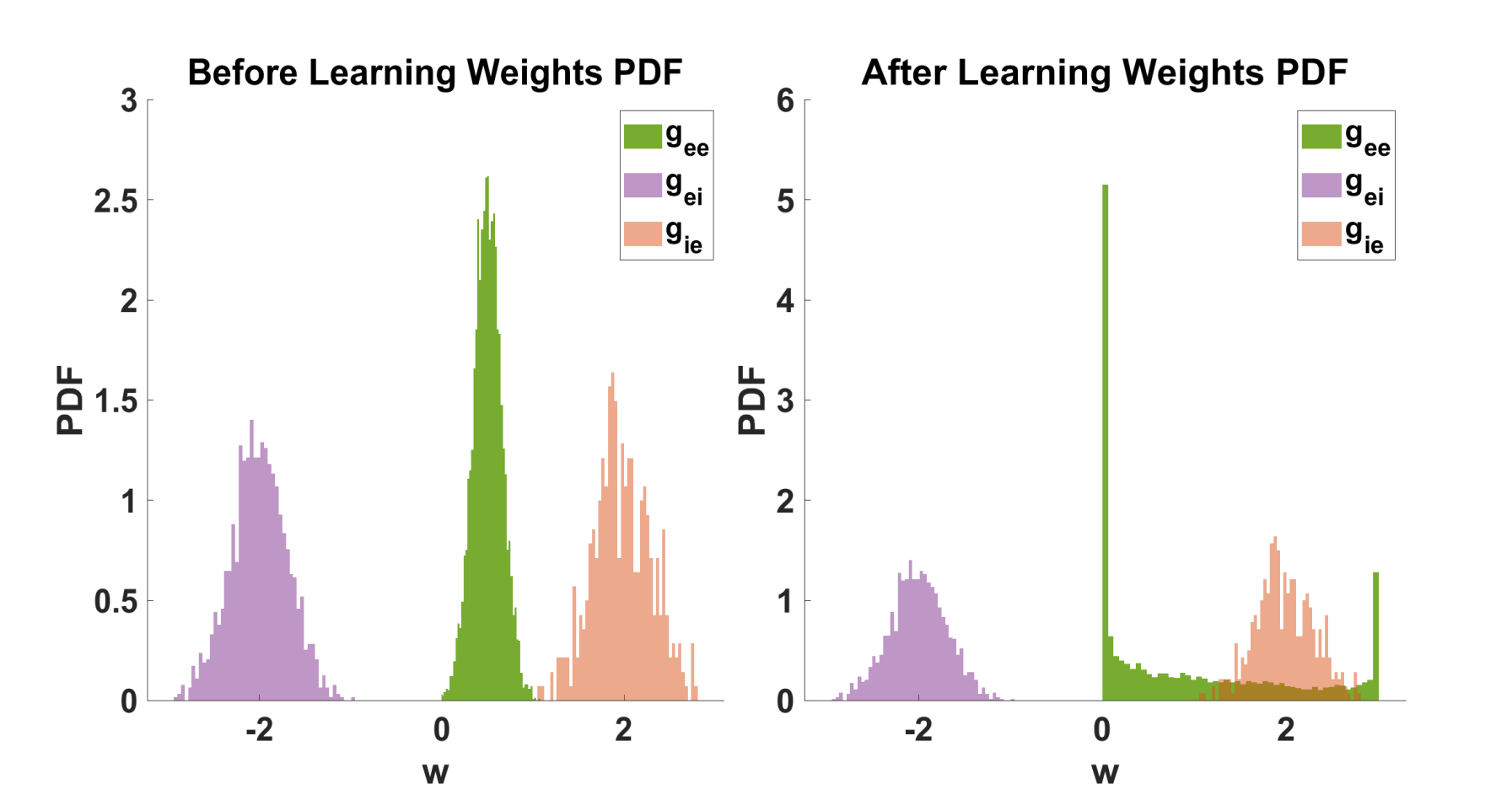


Left: Order vector of cells across different trials. Indices in each trial are sorted based on time to first spike. Color coding is based on last trial which this order stabilizes. **Right:** Correlation matrix of order vectors. Evidence for stability of order (our memory) after trial 400.

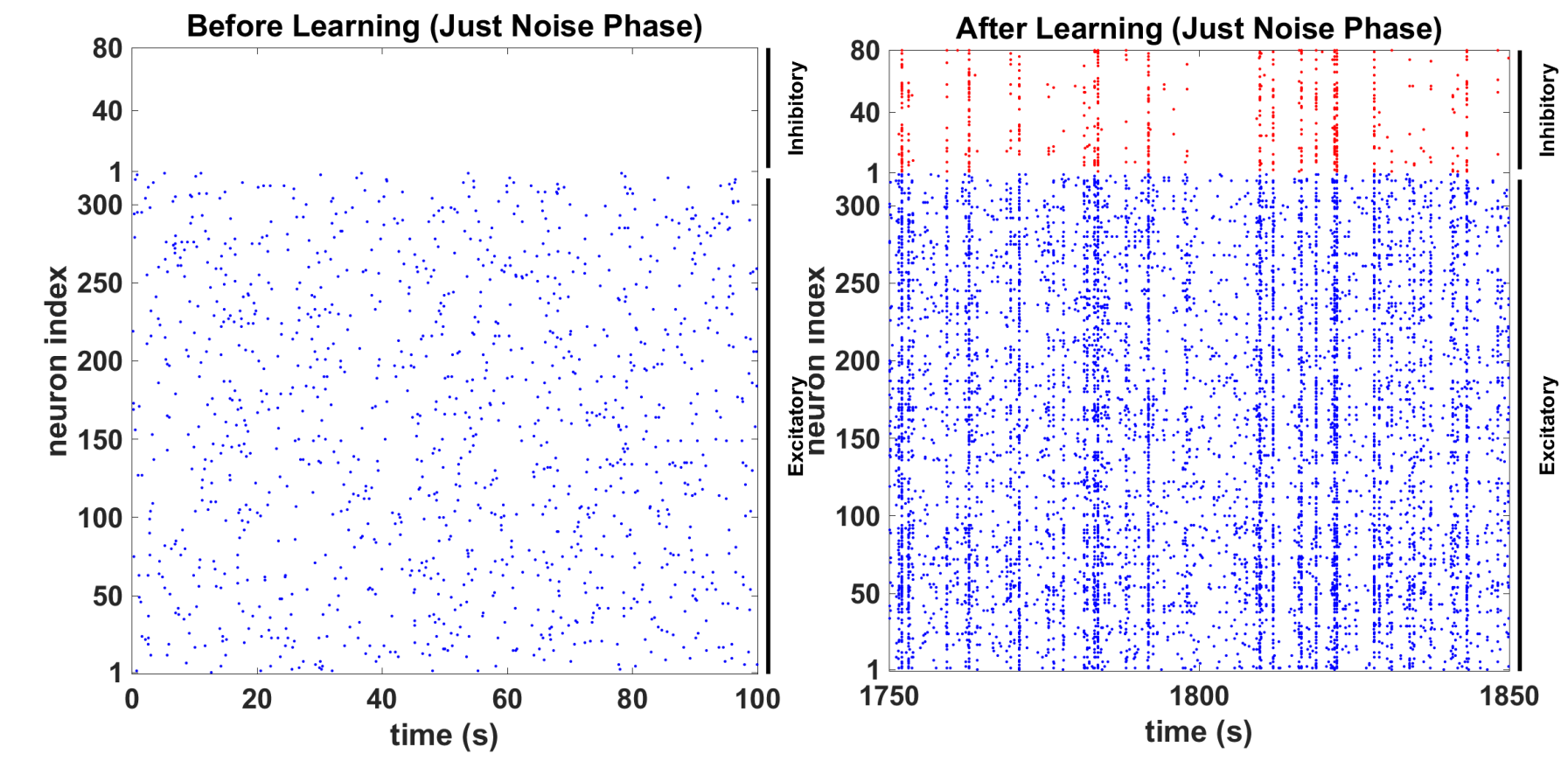
Imprinting Two Memories



Pre and Post Learning



STDP rule evolving synaptic weights (just g_{ee}) in time. **Left:** Initial histogram of synaptic weights before learning. **Right:** Histogram of synaptic weights at the end of the learning phase.



Raster plots of spontaneous network activity (driven by noise). **Left:** Noise driven activity of the network before learning. **Right:** Noise driven activity of the network after learning showing the presence of network bursts.

Conclusions

The results of this research demonstrate that it is feasible to construct a biologically inspired spiking neural network capable of storing memories in a manner similar to classical Hopfield networks.

Our findings indicate that such networks can stabilize and maintain memory patterns despite the presence of noise, which is a significant challenge in neural network models.

Future Work

- Decoder Development:** Build a decoder population of neurons to read the current network activity and identify active memories online.
- Heterogeneity:** Introduce parameters variability in the Izhikevich model to better describe actual neural circuits, e.g., cortex or hippocampus.
- Memory Capacity:** Quantify the network memory storage and retrieval capabilities.
- Feature Analysis:** Examine additional features beyond time to first spike ordering of learned activity which could be timing of other neuronal spikes after the first one.
- Plasticity Mechanisms:** incorporate additional plasticity mechanisms, including synaptic scaling and more detailed STDP rules. Also apply plasticity to the other two g_{ei} and g_{ie} connections.

References

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- M. C. W. Van Rossum MC, Bi GQ, Turrigiano GG. Stable Hebbian learning from spike timing-dependent plasticity. *Journal of Neuroscience*, 2000
- D. V. Buonomano, A learning rule for the emergence of stable dynamics and timing in recurrent networks. *Journal of Neurophysiology*, 2005
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